

SUBIN PULARI

Email: spulari@hse.ru ◇ Alternate Email: gsubing@gmail.com

Homepage: <https://pularisubin.github.io/> ◇ ORCID: <https://orcid.org/0000-0001-8426-4326>

PROFESSIONAL EXPERIENCE

Assistant Professor

September 2025 – Present

Faculty of Computer Science, HSE University, Moscow

CNRS Postdoctoral Researcher

October 2024 – September 2025

LaBRI, Université de Bordeaux

Senior Project Associate

June 2024 – September 2024

Department of Computer Science and Engineering, Indian Institute of Technology Kanpur

EDUCATION

Indian Institute of Technology Kanpur

July 2018 – June 2024

Ph.D. & Master of Technology (M.Tech) in Computer Science and Engineering

CPI: 10/10

Department of Computer Science and Engineering

Advisors: Dr. Satyadev Nandakumar and Dr. Sunil Simon

National Institute of Technology Calicut

July 2014 – May 2018

Bachelor of Technology (B.Tech) in Computer Science and Engineering

CGPA: 9.53/10

Department of Computer Science and Engineering

RESEARCH INTERESTS

- Algorithmic Information Theory.
- Computational Complexity Theory.
- Meta-complexity and Pseudorandomness.
- Computability and Complexity in Analysis.
- Ergodic Theory and Symbolic Dynamical Systems.
- Effective Fractal Dimension and Finite-State Dimension.

PUBLICATIONS

1. **One-Way Functions and Polynomial Time Dimension** (with Satyadev Nandakumar, Akhil S and Suranjona Sarma), *The 30th International Conference on Randomization and Computation, RANDOM 2026, Boston, USA (Accepted for Publication)*.
2. **Finite-State Dimension for Continued Fractions: Betting, Entropy and Normality** (with Satyadev Nandakumar and Akhil S), *51st International Symposium on Mathematical Foundations of Computer Science (MFCS 2026), Paris, France (Accepted for Publication)*.
3. **On Normality and Equidistribution for Separator Enumerators**, *Computability in Europe (CiE 2026), Trier, Germany (Accepted for Publication)*.
4. **Randomness Extraction Fails for Finite-State Dimension** (with Akhil S), *41st Annual Symposium on Logic in Computer Science (LICS 2026), Lisbon, Portugal (Accepted for Publication)*.

5. **A Markov-Chain Characterization of Finite-State Dimension and a Generalization of Agafonov’s Theorem** (with Laurent Bienvenu and Hugo Gimbert), *17th Latin American Theoretical Informatics Symposium (LATIN 2026)*, Florianópolis, Brazil (Accepted for Publication, Presented at the Conference).
6. **The Agafonov and Schnorr–Stimm Theorems for Probabilistic Automata** (with Laurent Bienvenu and Hugo Gimbert), *45th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2025)*, Goa, India.
7. **Real Numbers Equally Compressible in Every Base** (with Satyadev Nandakumar), *ACM Transactions on Computation Theory*, 2025, Volume 17, Issue 3, Article No.: 16, Pages 1–28.
8. **Point-to-Set Principle and Constructive Dimension Faithfulness** (with Satyadev Nandakumar and Akhil S), *49th International Symposium on Mathematical Foundations of Computer Science (MFCS 2024)*, Bratislava, Slovakia.
9. **Finite-State Relative Dimension, Dimensions of A.P. Subsequences and a Finite-State van Lambalgen’s Theorem** (with Satyadev Nandakumar and Akhil S), *Information and Computation*, Volume 298, June 2024, 105156.
10. **A Weyl Criterion for Finite-State Dimension and Applications** (with Jack H. Lutz and Satyadev Nandakumar), *48th International Symposium on Mathematical Foundations of Computer Science (MFCS 2023)*, Bordeaux, France.
11. **Real Numbers Equally Compressible in Every Base** (with Satyadev Nandakumar), *40th International Symposium on Theoretical Aspects of Computer Science (STACS 2023)*, Hamburg, Germany, 2023 (Invited to the STACS special issue of *ACM Transactions on Computation Theory*).
12. **Finite-State Relative Dimension and the Dimensions of A.P. Subsequences** (with Satyadev Nandakumar and Akhil S), *17th Annual Conference on Theory and Applications of Models of Computation (TAMC 2022)*, Tianjin, China.
13. **Ergodic Theorems and Converses for PSPACE Functions** (with Satyadev Nandakumar), *Theory of Computing Systems*, 2022.
14. **Ergodic Theorems for PSPACE Functions and Their Converses** (with Satyadev Nandakumar), *46th International Symposium on the Mathematical Foundations of Computer Science (MFCS 2021)*, Tallinn, Estonia.
15. **An Analogue of Pillai’s Theorem for Continued Fraction Normality and an Application to Subsequences** (with Satyadev Nandakumar, Prateek Vishnoi and Gopal Viswanathan), *Bulletin of the London Mathematical Society*, Volume 53, Issue 5, October 2021, Pages 1414–1428.

PREPRINTS

1. **Efficient Constructions of Finite-State Independent Normal Pairs**, *arXiv: 2602.23030*.

RESEARCH VISITS

Tata Institute of Fundamental Research, Mumbai
Visiting Students’ Research Programme (VSRP 2018)

May 2018 – July 2018

- Research supervised by Dr. Piyush Srivastava, Reader, School of Technology and Computer Science, TIFR, Mumbai.

Indian Institute of Technology Kanpur
Students-Undergraduate Research Graduate Excellence (SURGE 2017)

May 2017 – July 2017

- Research supervised by Dr. Satyadev Nandakumar, Associate Professor, CSED, IITK.

WORKSHOPS ATTENDED

Complexity as a kaleidoscope

March-April 2025

- Centre International de Rencontres Mathématiques (CIRM) in Marseille, France.

IMS Graduate Summer School in Logic

July 2022

- Summer School in Mathematical Logic at Institute of Mathematical Sciences (IMS), National University of Singapore.

CSA50 - Pratiksha Trust Workshop on Theoretical Computer Science

January 2019

- Venue: Indian Institute of Science, Bangalore, India.

TEACHING EXPERIENCE

HSE University, Moscow

2025 – 2026

Classroom Teaching

- Mathematical Analysis 2 (2025)
- Theory of Computation / Computational Complexity Theory (2025–2026)
- Algebra (2026)

Indian Institute of Technology Kanpur

2018 – 2022

Graduate Teaching Assistantship

- Principles of Programming Languages (2018 and 2019)
- Computer Organization (2018)
- Algorithmic Information Theory (2019)
- Theory of Computation (2020)
- Mathematics for Computer Science (2020 and 2021)
- Computational Complexity (2021)
- Quantum Computing (2022)

STUDENT SUPERVISION

Undergraduate

HSE University, Moscow

- Zabruskov Philip Kirillovich (2026), *Studies on the Normality-Dimension Thesis*.

INVITED TALKS

1. **Mathematical Seminar, Faculty of Computer Science, HSE Moscow (13 February 2026)**, *When Normality Meets Automata: Dimension, Selection, and Randomization*.
2. **Séminaire Algorithmique, Groupe de recherche en informatique, image et instrumentation de Caen (GREYC), Caen (11 March 2025)**, *A Duality Between One-Way Functions and Non-Robustness of Polynomial Time Dimension*.
3. **Logic Seminar, Department of Mathematics, National University of Singapore (5 March 2025)**, *On the Compressibility of Real Numbers: New insights using exponential sums*.
4. **Automates Seminar, Institut de Recherche en Informatique Fondamentale (IRIF), Paris (17 January 2025)**, *On the Compressibility of Real Numbers: New insights using exponential sums*.

SEMINAR AND CONFERENCE TALKS

1. **A Markov-Chain Characterization of Finite-State Dimension and a Generalization of Agafonov's Theorem**, *17th Latin American Theoretical Informatics Symposium (LATIN 2026)*, Florianópolis, Brazil (13 April 2026).
2. **The Agafonov and Schnorr–Stimm Theorems for Probabilistic Automata**, *45th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS)*, Goa, India (17 December 2025).
3. **The Normality–Dimension Thesis: Recent Progress and Future Directions**, *CCR 2025: Computability, Complexity and Randomness & Journées Calculabilités*, Bordeaux, France (19 June 2025).
4. **On the Compressibility of Real Numbers: Certain insights using Fourier analytic methods**, *M2F Seminar, LaBRI, Université de Bordeaux* (12 November 2024).
5. **A Weyl Criterion for Finite-State Dimension and Applications**, *48th International Symposium on Mathematical Foundations of Computer Science (MFCS 2023)*, Bordeaux, France (29 August 2023).
6. **Real Numbers Equally Compressible in Every Base**, *40th International Symposium on Theoretical Aspects of Computer Science (STACS 2023)*, Hamburg, Germany (9 March 2023).

REPORTS / DISSERTATIONS

On Resource Bounded Ergodic Theorems and the Utility of Exponential Sums in Algorithmic Information Theory

PhD Thesis (IIT Kanpur)

- Supervised by Dr. Satyadev Nandakumar and Dr. Sunil Simon.

On Certain Applications of Fourier Analysis in the Theory of Finite-State Dimension

M.Tech Thesis (IIT Kanpur)

- Supervised by Dr. Satyadev Nandakumar and Dr. Sunil Simon.

On Finite State Ergodic Markov Chains

B.Tech Major Project (NIT Calicut)

- Supervised by Dr. K. Murali Krishnan.

ACADEMIC HONORS AND AWARDS

- Gold Medal for the highest CGPA in Bachelor of Technology (B.Tech) Computer Science and Engineering 2018, National Institute of Technology Calicut.
- Best Student Project Award 2018 by Tata Consultancy Services Limited (TCS) for the B.Tech project titled *On Finite State Ergodic Markov Chains*.
- Best Project Award in B.Tech Computer Science and Engineering 2018 by C.R.E.C. Parent Teacher Association, National Institute of Technology Calicut for the B.Tech project titled *On Finite State Ergodic Markov Chains*.

OTHER PROFESSIONAL SERVICES

- **Conferences refereed** – Computability in Europe (CiE), Computational Complexity Conference (CCC), Mathematical Foundations of Computer Science (MFCS).

REFERENCES

Available on request.